

Network Science

Lecture 05: Social Context and Link Formation

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Speaker notes

Lectures 03 and 04 treated the graph as a structure of anonymous nodes and unlabeled edges — the only question was "is the edge there?". This lecture adds context: nodes get *attributes* (age, track, beliefs), edges get *timestamps*, and the graph gets *foci* (shared social settings). The central tension shifts from the **computational** gap of L03–L04 (what we want to compute is NP-hard) to a **causal** gap: the mechanism behind a network pattern — selection vs. socialization — cannot be identified from a single snapshot. The fix is not a better algorithm but richer data: longitudinal observations and explicit social contexts.



From Structure to Context

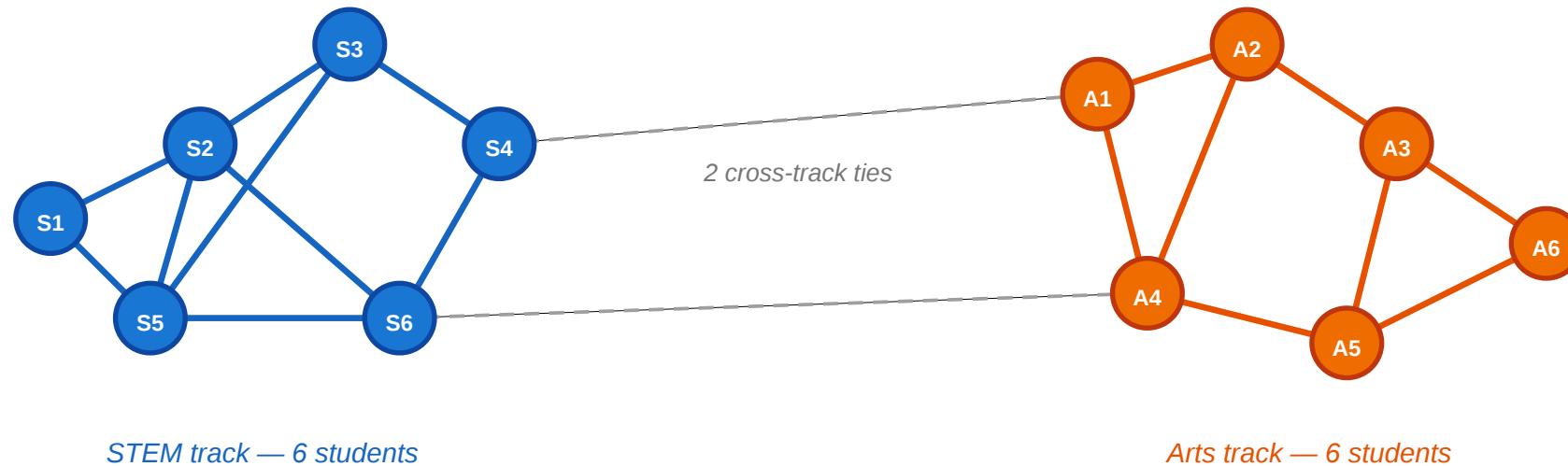
In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

- ① Is the within-track clustering more than we'd expect by chance?
- ② Did students choose friends from the same track, or did they become more alike after befriending?
- ③ Could shared classes (foci) explain the clustering without any personal preference?
- ④ Would mild preferences suffice to create this pattern?

Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

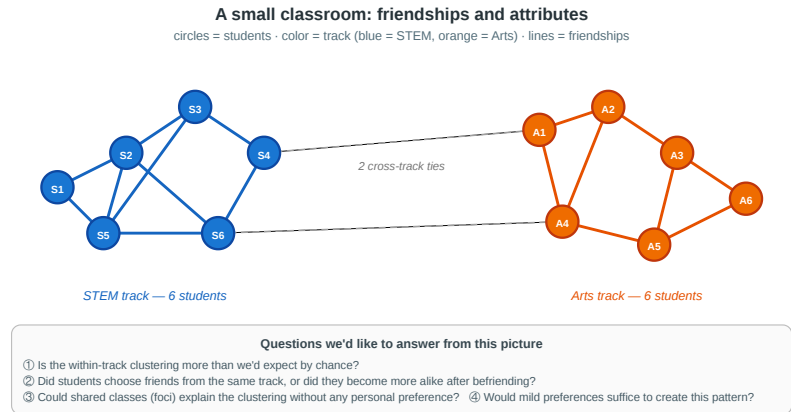
Speaker notes

This figure is the anchor for the whole lecture. Use it to provoke the four questions on the next slide before any definition arrives. By the end of the session students should be able to articulate four different explanations for the visible pattern and say which data would distinguish them.



Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
 2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
 3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
 4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?
- Every concept in today's lecture answers one of these.



Speaker notes

Each question maps to a module: (1) homophily and its formal baseline, (2) selection vs. socialization and the causal identification problem, (3) affiliation networks and focal closure, (4) online link formation and (5) Schelling segregation as a model for how mild local preferences generate sharp global patterns. Keep referring back to the classroom figure throughout.



A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)



Speaker notes

This slide is the L05 analogue of L03's "theory vs. measurement" and L04's roadmap table. The new axis is causal identification rather than computational tractability. The rest of the lecture fills in the four rows.



Takeaway

L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

This frames the shift cleanly: from computation to causation. Refer back to this framing when each module introduces its specific gap.

Homophily

Guiding Question: Do people form ties because they are similar — and how would we know?

Speaker notes

The key point is that homophily is a *statistical* pattern requiring a counterfactual baseline. "Many within-group ties" is not evidence of homophily unless we know how many ties we would expect under random mixing. Students should leave this module knowing that declaring homophily requires a baseline, not just a visual impression.



What Does Homophily Look Like?

Homophily as similarity-biased tie formation



Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Homophily = tie formation is biased toward similarity. Within-group edges (colored) dominate; cross-group edges (gray) are rare. The pattern looks clear-cut — but a strong within-group share could also arise mechanically when one class is much larger than the other. The next slide makes that worry precise.

Speaker notes

This is the visual hook for the whole homophily module. Students see the pattern first — clustered, segregated, easy to spot — then in the next slide they discover that "easy to spot" is not the same as "more than chance." Use this slide to land the everyday meaning of homophily ("birds of a feather flock together") before any formula appears.

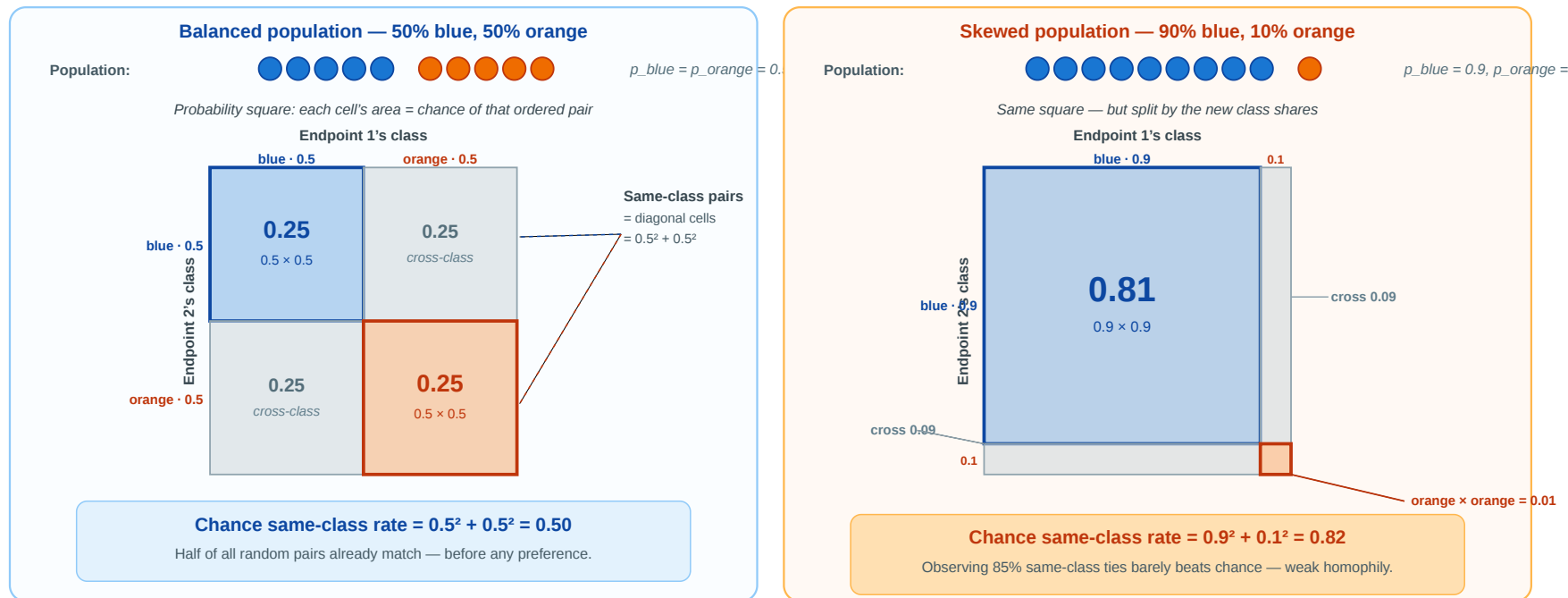


Homophily — What Counts as "More Than Chance"?

Before calling that pattern *homophily*, we have to ask: **how many same-class ties would we see if people paired up purely at random?** Only the *excess* over that chance rate is evidence of a preference for similarity.

If we ignored preferences and paired nodes at random — how often would the two endpoints share a class?

This "chance rate" is what an observed within-group rate must beat to count as homophily. The probability square shows the rate in two populations.



Each cell's area is the chance of that ordered pair; the **colored diagonal cells** are the same-class outcomes. In the skewed population (right), 82% of random pairs already match — so an observed 85% within-group rate is almost noise. Next slide: we name this chance rate H_{base} .

Speaker notes

This slide opens with the *question* — what would chance alone produce? — not with a formula. The probability-square visualisation lets students see the answer: each cell is an ordered pair of endpoints, its area is the joint probability of that pair, and the diagonal cells are the same-class outcomes. The skewed 90/10 panel is the punchline: chance alone explains 82% within-group ties, so any honest claim of homophily must beat that rate. Only on the next slide do we attach the name H_{base} and the formula $\sum p_i^2$ — by that point students already understand what the formula is doing geometrically: summing the areas of the diagonal cells.

Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x . Formally, for a categorical attribute with class shares $p_1, \dots, p_k$:

- **Random-mixing baseline** (area of the diagonal cells in the probability square):

$$H_{\text{base}} = \sum_{i=1}^k p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs} .
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

With the probability-square intuition from the previous slide, the formula $H_{\text{base}} = \sum p_i^2$ is just the area of the diagonal cells. The homophily index r is a variant of Newman's assortativity coefficient for categorical attributes (Newman, 2003); it is the standard way to quantify homophily in empirical network studies. The normalization by $(1 - H_{\text{base}})$ matters precisely because of the 90/10 case shown in the previous figure: reporting "85% same-group" without the baseline would overstate the evidence for homophily.

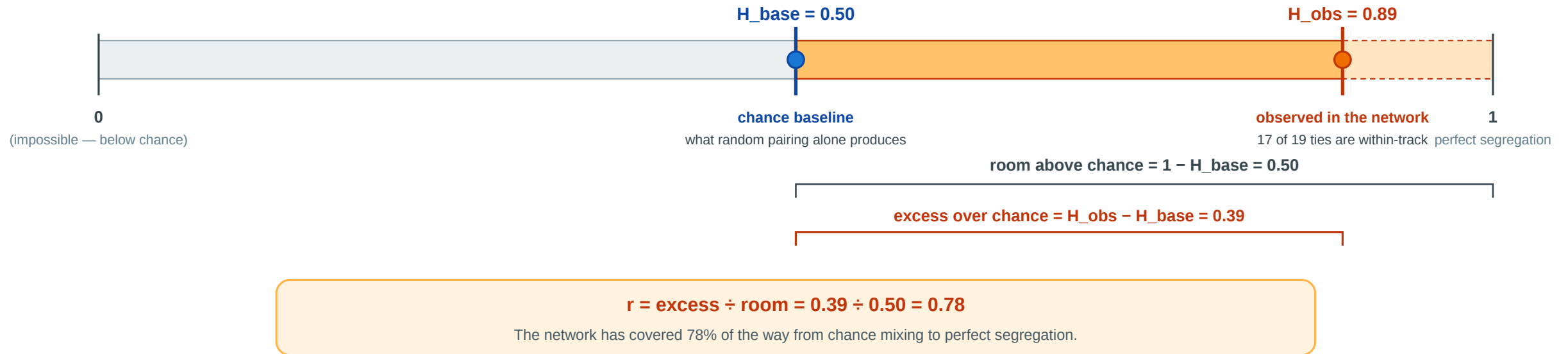
Homophily — Three Numbers, One Story

Measure	What it tells you	Depends on
H_{base}	The chance rate. What random pairing alone would produce. Large when one class dominates the population.	Class shares only
H_{obs}	What the data actually shows. Count the within-class edges and divide by the total.	Observed edges only
r	How far the network has moved from chance toward perfect segregation , on a 0–1 scale. $r = 0$: chance. $r = 1$: perfect segregation. $r < 0$: cross-class preference (heterophily).	Both



How far has the network moved from chance toward perfect segregation?

The homophily index r is the fractional position of the observed rate on the “room above chance”.



The classroom example: $H_{\text{base}} = 0.50$, $H_{\text{obs}} = 0.89$. The observed rate has covered **78% of the available room** above chance — that is the homophily index r .

Speaker notes

This slide gives each of the three measures a one-sentence plain-English meaning and shows r as a *fractional position* on the line between chance and perfect segregation. The visual is meant to dispel a common confusion: r is **not** $H_{\text{obs}} - H_{\text{base}}$ (the raw excess), but the *fraction of available room* the network has used. The "available room" is whatever lies above the chance baseline; in a balanced population the room is large (0.50), in a skewed 90/10 population the room is small (0.18). That denominator is what makes r comparable across populations with different class shares.

Homophily Baseline Trap

The **same observed within-group share** can mean very different things depending on the population composition. The table below holds $H_{\text{obs}} = 0.85$ fixed and varies the class shares:

Population shares	H_{base}	Room above chance	r	What you can honestly say
50% / 50%	0.50	0.50	0.70	Strong excess — 70% of the way to perfect segregation
70% / 30%	0.58	0.42	0.64	Substantial — 64% of available room used
90% / 10%	0.82	0.18	0.17	Barely above chance — almost nothing to explain
99% / 1%	0.98	0.02	< 0	<i>Below</i> chance — actually heterophilic!

Why this trap is so common. Journalists and researchers often report "X% of friendships are within-group" *without the baseline*. In a setting that is 80% one type and 20% another, H_{base} is already 0.68 — so an "82% within-group" headline is barely homophily. The trap appears whenever group sizes are imbalanced: school ethnicity studies, workplace gender networks, online platform demographics. The "Birds of a Feather" review (McPherson, Smith-Lovin & Cook 2001) insists on reporting effect sizes *relative to* a population-specific baseline for exactly this reason.

Never interpret "X% of ties are within-group" without asking: **X% compared to what baseline?** The baseline depends on the population, not on what feels intuitively high.

This is a useful didactic pause because students often read a high within-group percentage as direct evidence of strong preference. The expanded table shows that the same 0.85 within-group rate maps to $r = 0.70$ in a balanced population but to $r = 0.17$ — or even *heterophily* — in skewed populations. A concrete teaching example: a school that is 80% one ethnic group and 20% another will *mechanically* produce ~68% same-ethnicity friendships even under fully random mixing, before any preference. Reporting "82% same-group" out of context would overstate the evidence by a wide margin. This is precisely why the McPherson, Smith-Lovin & Cook (2001) review of homophily research demands a baseline.

Back to the Classroom

Both tracks have 6 students, so $p_{\text{STEM}} = p_{\text{Arts}} = 0.5$.

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- 17 within-track edges of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

Walk through the arithmetic side-by-side with the figure: point at the within-track clusters as you read off H_{obs} , then point at the two dashed cross-track ties as the remainder. The punchline — "strong homophily, but the mechanism is unknown" — is the bridge to Module 2. The same $r = 0.78$ would arise under pure selection, pure socialization, or pure focal closure. Distinguishing them requires more than a cross-sectional snapshot.

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$r = 0.78$ is strong — far above random mixing. But it does not tell us *why*. Is it selection, socialization, or shared context?

Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. Compute the random-mixing baseline H_{base} .
2. Compute the homophily index r .
3. Is r strong evidence of homophily? What could make it misleading?

Give them 4 minutes before revealing the fragments. (1) $H_{\text{base}} = 0.6^2 + 0.4^2 = 0.52$. (2) $r = (0.78 - 0.52)/(1 - 0.52) = 0.26/0.48 \approx 0.54$. (3) $r = 0.54$ is moderate-to-strong, but it can be misleading if the tracks share different physical spaces (focal closure), if students were assigned to tracks based on pre-existing friend groups (reverse causation), or if family background causes both track choice and friendship patterns (confounding). Reveal the three answer fragments one at a time as students suggest theirs.

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1. $H_{\text{base}} = 0.6^2 + 0.4^2 = 0.36 + 0.16 = 0.52$

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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference);
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation);
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding).



Takeaway

Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

Echo Chambers — The Hypothesis

Source: Cinelli et al. 2021

Claim	Online platforms expose users mostly to like-minded peers and sources.
Network signal	High homophily in the <i>interaction</i> network (likes, retweets, replies, comments) when nodes are coloured by political leaning.
Diffusion signal	Information circulates <i>within</i> groups rather than crossing them.

Why it's a homophily story. An echo chamber is not just "people who disagree online" — it is a region of the interaction graph where within-group ties dominate so heavily that the same content recirculates without leaving the cluster.

The empirical test (Cinelli et al. 2021, PNAS).

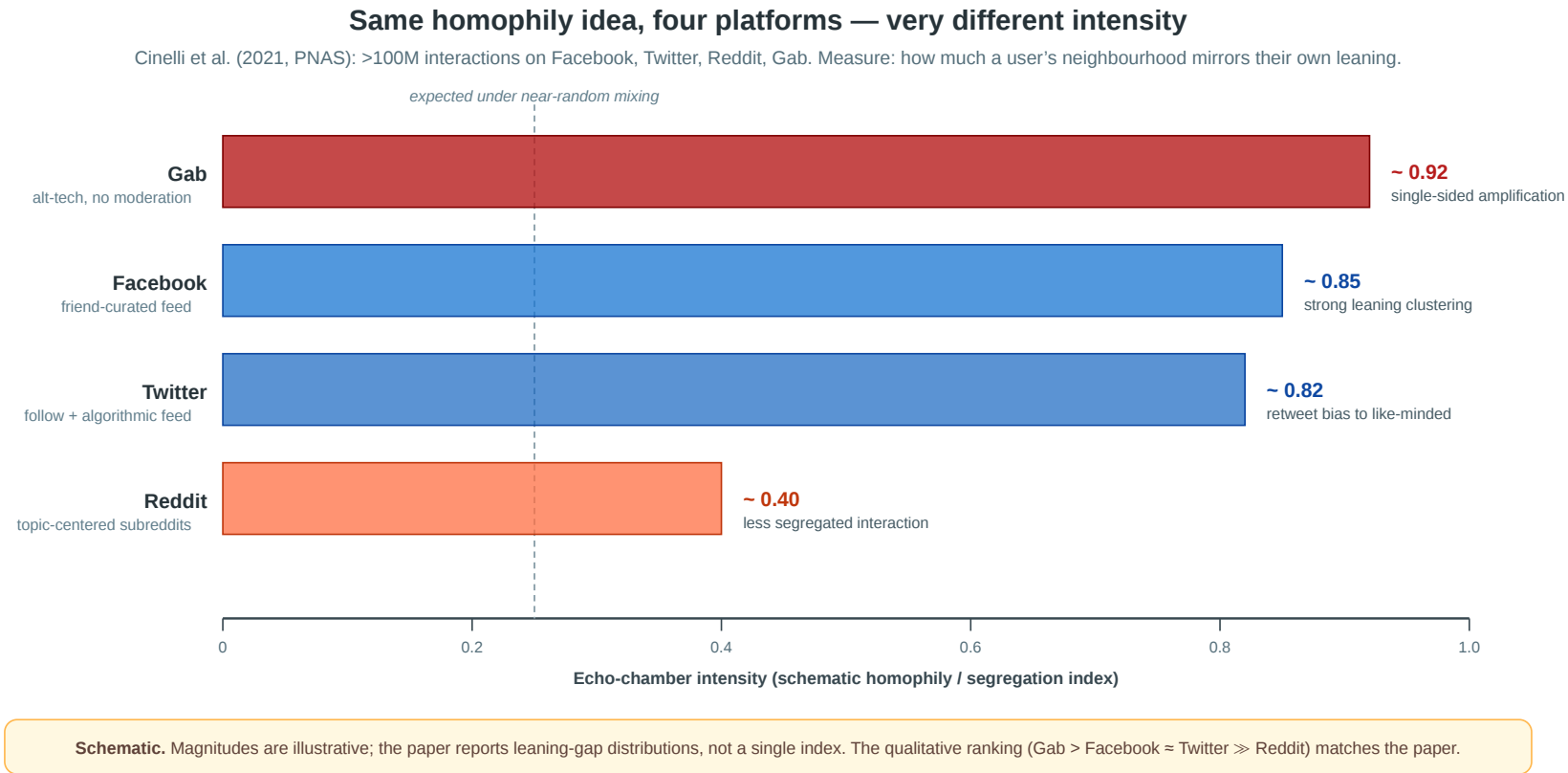
- Four platforms: **Facebook, Twitter, Reddit, Gab.**
- Over **100 million** interactions, hundreds of news outlets coded by political leaning.
- For each user, compute the *leaning gap* between the user and the people / outlets they interact with.
- A small leaning gap = the user lives in an echo chamber.

Different platform architectures (algorithmic feed, friend graph, subreddit topic, alt-tech) let us see whether echo chambers are universal or **architecture-dependent**.

Speaker notes

This opening slide gives the *network* framing for the echo chamber concept: it is a regional, homophily-based pattern in the interaction graph. The Cinelli et al. study is the largest cross-platform test of that claim. They operationalize echo-chamber intensity as how closely a user's interaction neighbourhood mirrors the user's own leaning. The four-platform design is what makes the study informative beyond Facebook — it lets the analyst distinguish "echo chambers exist online" from "echo chambers depend on how the platform is built."

Echo Chambers — Same Idea, Four Platforms

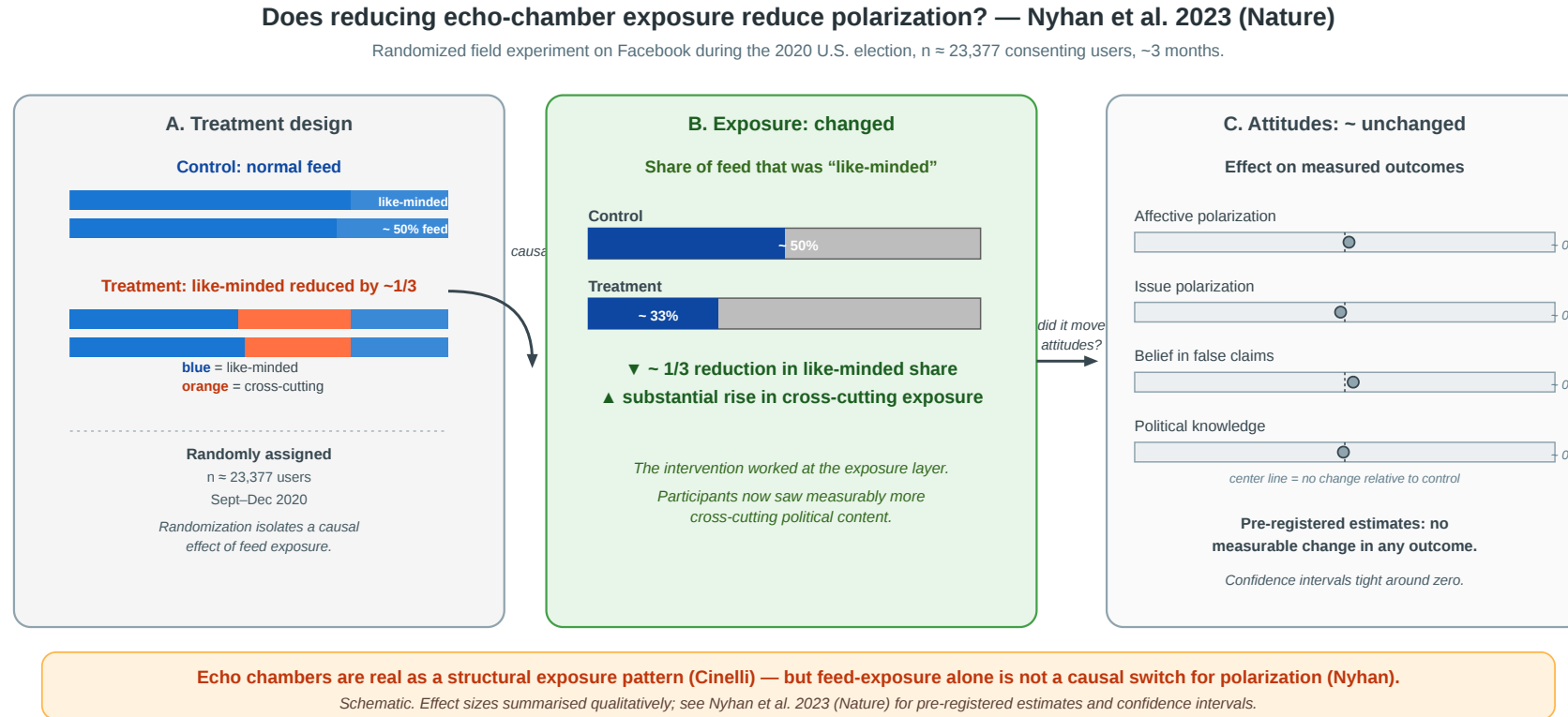


Architecture matters. Facebook and Twitter (friend / follow graphs amplified by ranking) show strong homophilic clustering by political leaning. **Gab** — small, unmoderated, single-sided audience — is even more extreme. **Reddit** is much less segregated: topic-centered subreddits force users to interact across leanings within the same subforum. The same homophily concept produces dramatically different intensities across designs.

The qualitative ranking (Gab > Facebook $\approx$ Twitter $\gg$ Reddit) is the central empirical result. Use the figure to make two pedagogical points: (i) "echo chamber" is not a binary — it is a measurable intensity along the homophily continuum from $r = 0$ to $r = 1$; (ii) the intensity is a *property of the platform's architecture*, not a fixed feature of human behaviour. Friend-graph platforms with ranking-based feeds make echo chambers stronger; topic-graph platforms make them weaker. This connects directly to the affiliation-network module that follows: subreddits are explicit foci, friend feeds are not.

Echo Chambers — Does Exposure Cause Polarization?

Source: Nyhan et al. 2023



Cinelli (2021) shows echo chambers exist as a structural pattern. Nyhan et al. (2023, Nature) ask whether that exposure *causes* polarization. In a pre-registered randomized field experiment around the 2020 U.S. election ($n \approx 23,377$ consenting Facebook users), they reduced exposure to like-minded sources by about one-third for ~3 months. Cross-cutting exposure rose substantially — but **affective polarization, issue polarization, false beliefs, and political knowledge did not measurably change** (estimates tight around zero).

Structure ≠ causation. Dense homophily can *reinforce* information, but a few weeks of changed feed exposure may be too weak, too late, or too indirect to move stable political attitudes. The network concept ("echo chamber as homophilic region") is clearer than any causal claim about its effects.

This is the closing slide for the homophily module's empirical case study, and it is also the moment to introduce the *structure vs. causation* distinction that the next module (selection vs. socialization) will generalize. The Nyhan study is methodologically the gold-standard test: it is randomized, pre-registered, well-powered, and uses the platform's own logs to verify that the treatment actually moved exposure. The null finding on attitudes is *not* evidence that echo chambers don't matter — it is evidence that the simple causal story "more cross-cutting exposure → less polarization" is too coarse. Possible reasons: pre-existing attitudes are stable on a multi-week scale; cross-cutting content may be ignored or counter-reasoned; the relevant causal channel may be selection into platforms, not feed composition. The point for the lecture: a clean *structural* claim (homophily exists) and a clean *causal* claim (homophily causes polarization) are different statements, and they require different evidence.

Selection and Socialization

Guiding Question: How can we distinguish whether similarity caused the tie or the tie caused the similarity?

Speaker notes

This is arguably the most important conceptual module in the lecture. The three mechanisms — selection, socialization, and contextual correlation — produce the same cross-sectional pattern. Distinguishing them requires temporal data and careful causal reasoning. Without longitudinal observations, the $r = 0.78$ we just computed is compatible with any mixture of the three.

Selection and Socialization

Guiding Question: How can we distinguish whether similarity caused the tie or the tie caused the similarity?

homophily = selection + socialization + context

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

Speaker notes

All three mechanisms are consistent with a high homophily index. A cross-sectional snapshot — friendships and attributes measured at one point in time — cannot separate them. This is the causal wall that L05 is built around.



Same Snapshot, Three Stories

Suppose we observe: **STEM students mostly befriend STEM students.**

Mechanism	Concrete story	What data would help?
Selection	Students first choose STEM, then choose friends from the same track.	Friendships observed <i>after</i> track choice
Socialization	Friends influence each other to choose the same elective or track.	Track changes observed <i>after</i> friendships
Contextual correlation	STEM students share labs, schedules, and buildings, so they meet more often.	Timetables, room assignments, shared courses

The graph snapshot is identical in all three stories. The difference is the **time order** and the **unobserved context**.



Speaker notes

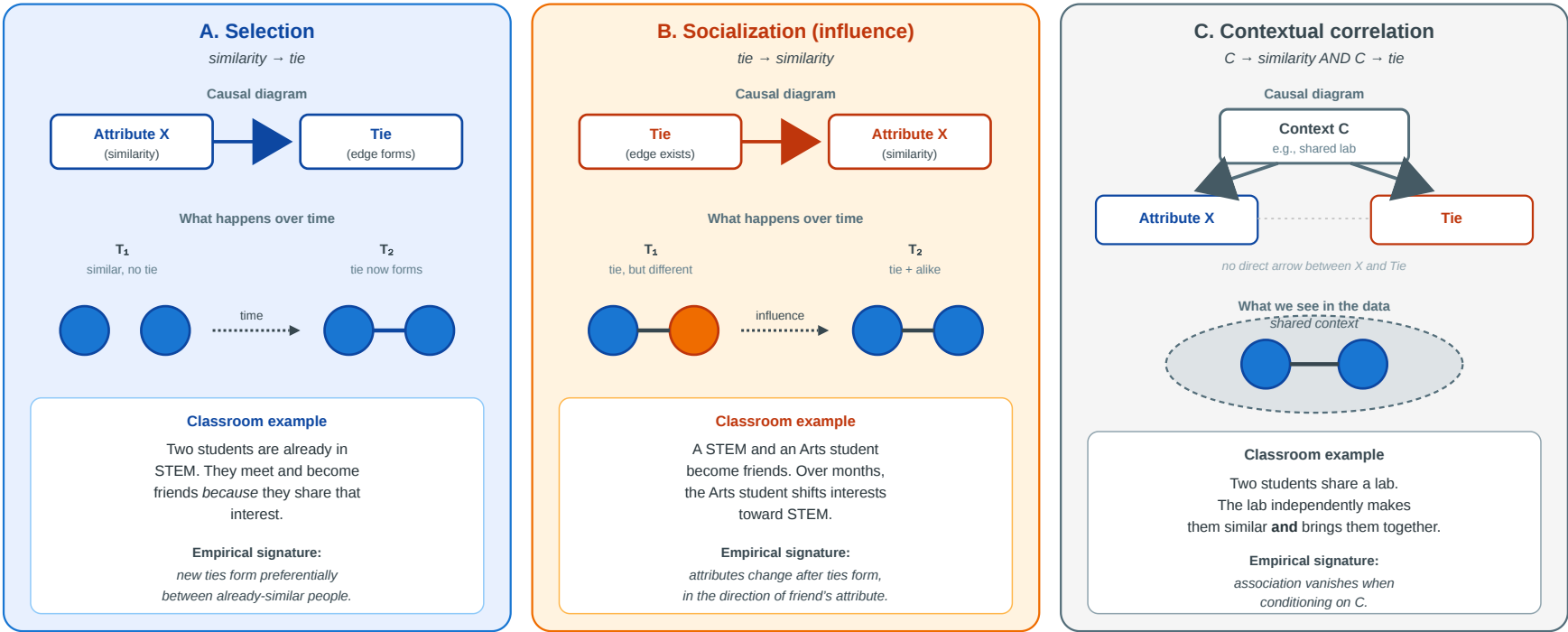
Use this slide to make the causal distinction tangible before showing the causal diagrams. The point is not that the mechanisms are mutually exclusive; in real schools all three can operate. The task is to ask which data would let us estimate their relative contribution.



Causal Diagrams

Three causal mechanisms — same observed pattern

All three produce the cross-sectional fact “connected nodes are similar.” Distinguishing them needs temporal data or knowledge of hidden context.



Speaker notes

Walk through the three causal diagrams: selection (similarity $\rightarrow$ tie), socialization (tie $\rightarrow$ similarity), context (third factor $\rightarrow$ both). Without temporal data, all three explain the same cross-sectional correlation. The point is not that one is always right; in real networks, all three usually operate simultaneously. The analyst's job is to measure their relative contributions.

Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, [New England Journal of Medicine 357, 370–379](#).

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

Speaker notes

The Framingham study is the canonical longitudinal analysis of selection vs. socialization. Its strength is the multi-decade time series: because both the network and the attribute (BMI) were measured repeatedly, the authors could observe the temporal order — who became obese *before* whom — and test whether the pattern fit selection, socialization, or confounding. The 57% figure is a relative risk increase among nominated friends compared to non-friends, controlling for geographic proximity.

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Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

What the Study Could Not See

- 1. Confounders.** Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.
- 2. Homophily in tie formation.** The study adjusted for *existing* attribute similarity but could not fully exclude the possibility that people selectively *formed* or *maintained* ties with others on similar weight trajectories — a dynamic form of selection invisible to static controls.
- 3. Missing network.** The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

Speaker notes

This slide is the L05 analogue of L03's "what Kossinets-Watts could not see" and L04's "what Zachary could not see". The pattern is deliberate: each lecture celebrates an empirical result and then identifies its blind spots. The Framingham critique is not a rejection of the study — it is a reminder that even 32-year longitudinal data with repeated measures cannot eliminate all causal alternatives. The lesson is humility, not nihilism.



Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Longitudinal data: friendships and smoking status at multiple time points. (2) Selection: new ties form between people who already both smoke. Socialization: non-smokers start smoking after befriending smokers. (3) Shared environment: dorm floor placement may independently cause both friendship formation and smoking uptake (e.g. shared common rooms where smoking happens).

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1. Longitudinal observations — friendships and smoking status at multiple time points, with newly formed ties and newly adopted behavior visible.

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2.

- *Selection*: new friendships disproportionately form between people who **already** share smoking status.
- *Socialization*: people who befriend smokers become more likely to **start** smoking in the next observation period.

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- *Selection*: new friendships disproportionately form between people who **already** share smoking status.
- *Socialization*: people who befriend smokers become more likely to **start** smoking in the next observation period.

3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.



Takeaway

Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

Affiliation Networks

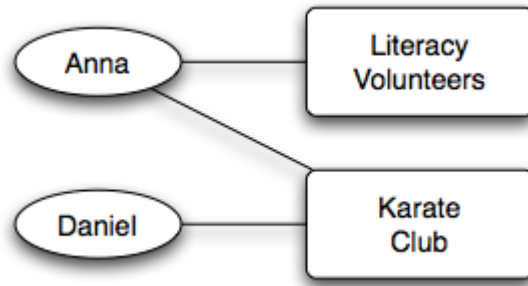
Guiding Question: How do we represent the social settings — classes, clubs, workplaces — in which ties form?

Speaker notes

Affiliation networks make the "shared context" mechanism from Module 2 explicit and formal. Instead of treating context as an unobserved confounder, we model it directly: people are connected to foci, and ties between people arise through their shared foci. This is the modeling answer to Question 3 from the opening slide.



Bipartite Affiliation Model



Source: Easley & Kleinberg 2010

An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

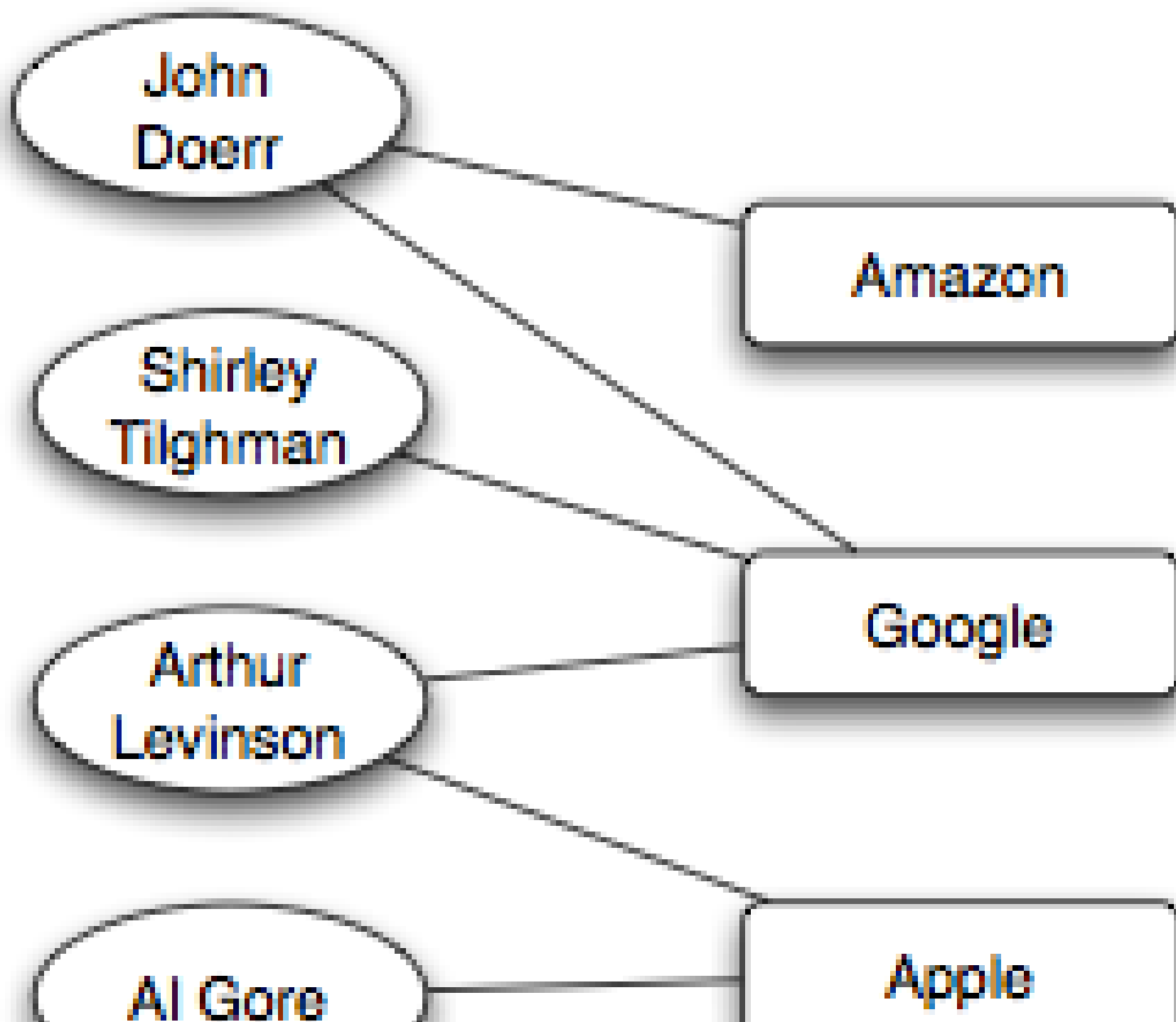
- P : actor nodes (persons)
- F : focus nodes (groups, locations, classes, organizations)
- E : edges only between P and F — "person p participates in focus f "

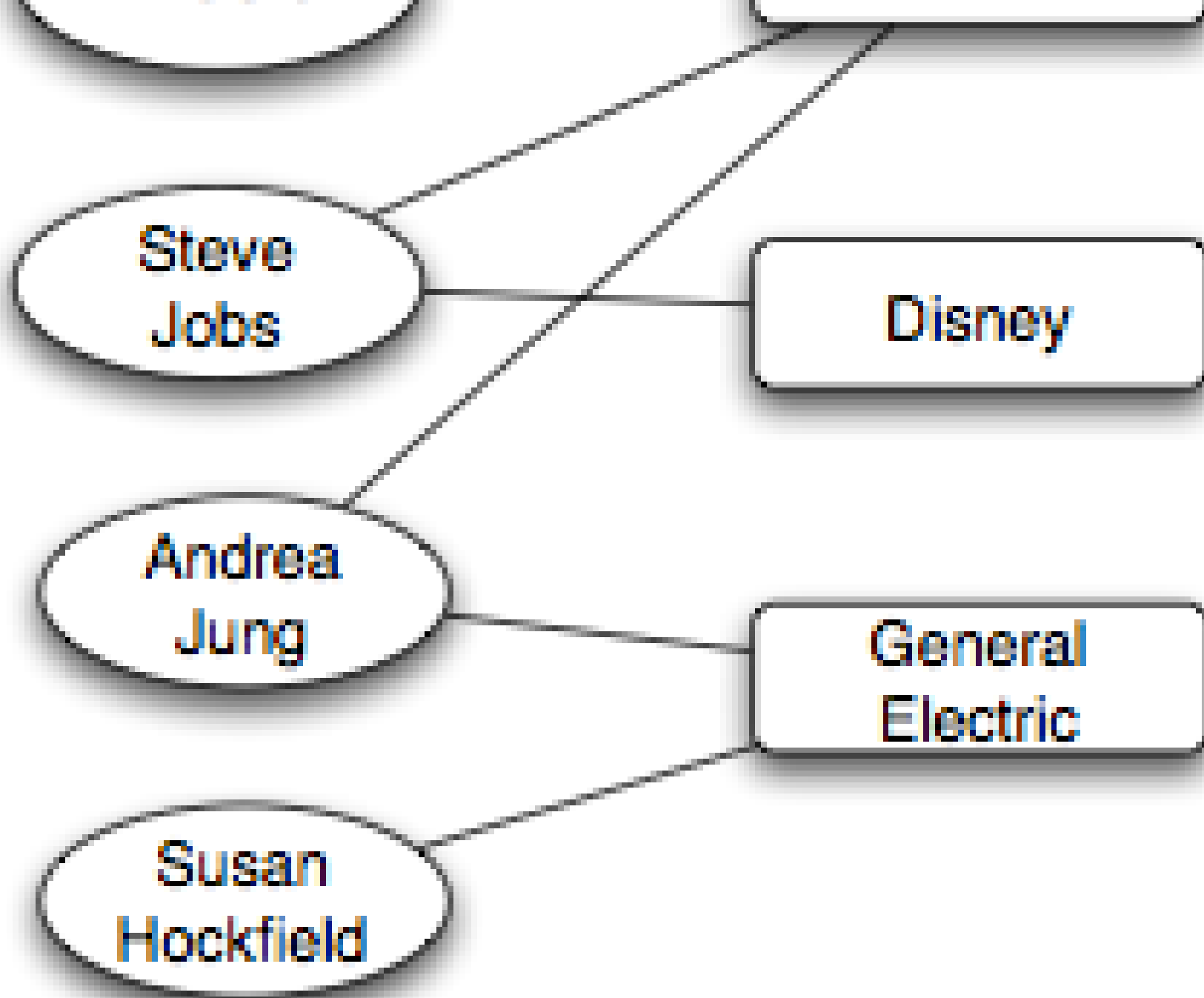
Two persons sharing a focus creates a *potential* for a person–person tie.

Speaker notes

This connects back to Lecture 02's bipartite graphs. The bipartite structure is a modeling decision: we choose to represent contexts as a separate node type rather than as edge attributes. The payoff is that closure processes become explicit and testable. The classroom scenario can be modeled this way: the two tracks are foci, and within-track friendships arise partly through shared-focus exposure.

Example — Corporate Board Interlocks





Reading the Board Network

An affiliation network is bipartite, but we usually want to ask one-sided questions about *people* or *firms*. The trick is to **project** the bipartite graph onto one side of the partition. There are two such projections, each answering a different question.

Projection	Nodes	Edge	Edge weight	Question it answers
Director network (person side)	People	Two directors sit on ≥ 1 shared board	Number of shared boards	Who can exchange career info / governance norms? Who is structurally similar?
Company network (focus side)	Firms	Two firms share ≥ 1 director	Number of shared directors	Where can strategy or practices diffuse? Which firms are strategically interlocked?

Toy example. Three directors d_1, d_2, d_3 on two boards b_1, b_2 ; $d_1, d_2 \in b_1$ and $d_2, d_3 \in b_2$.

- *Director projection:* $d_1!-!d_2, d_2!-!d_3 - d_2$ is a bridge.
- *Company projection:* $b_1!-!b_2$ — one edge, via the shared director d_2 .

The projection **destroys information**: once projected, a $d_1!-!d_2$ edge no longer tells you *which* board mediated it. You cannot recover the bipartite graph from either one-mode projection alone.

Three rules of thumb.

1. **A projection edge = opportunity for exposure**, not a friendship and not a confirmed information channel. A quarterly board meeting is a weak channel.
2. **Density rises sharply.** A board of size s contributes $\binom{s}{2}$ projected edges from a single focus — a 12-person board adds 66 director-director edges. Projections look denser than the underlying network.
3. **Use weights, not just edges.** Two directors sharing 5 boards is a much stronger signal than sharing 1 — but the unweighted projection makes them look identical.



The bipartite-to-one-mode projection is the key operational move when working with affiliation data. The choice of projection is *not* neutral: it foregrounds either person-level questions (career networks, peer exchange) or organization-level questions (strategic interlocks, diffusion across firms). Use the toy example on the slide to walk through the construction: pen and paper, three directors and two boards, then mechanically derive both projections. Emphasize the information-loss point: the director-director projection forgets *which* board mediated each tie. Practical consequences: (i) interpreting clustering coefficients in projections is tricky because boards create complete subgraphs (cliques) on their members; (ii) edge weights are almost always informative and should be reported. The "exposure not friendship" caveat is the same point as in the diffusion/contagion literature — co-membership is a *channel*, not yet a *cause*.

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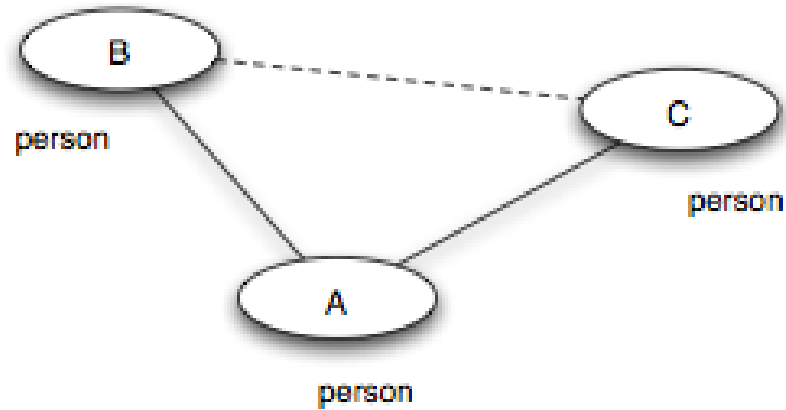
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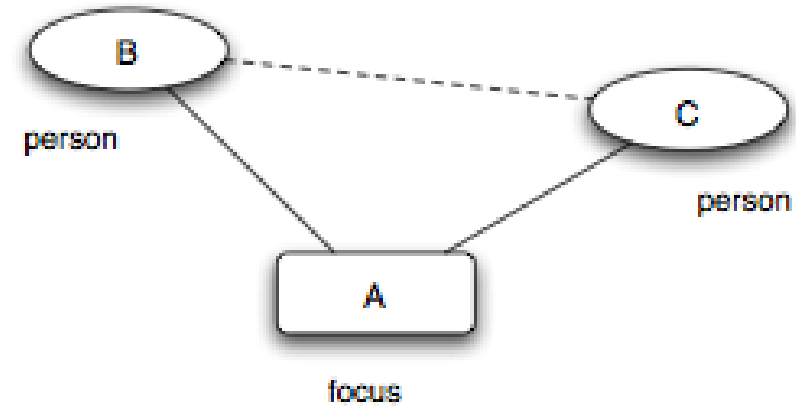
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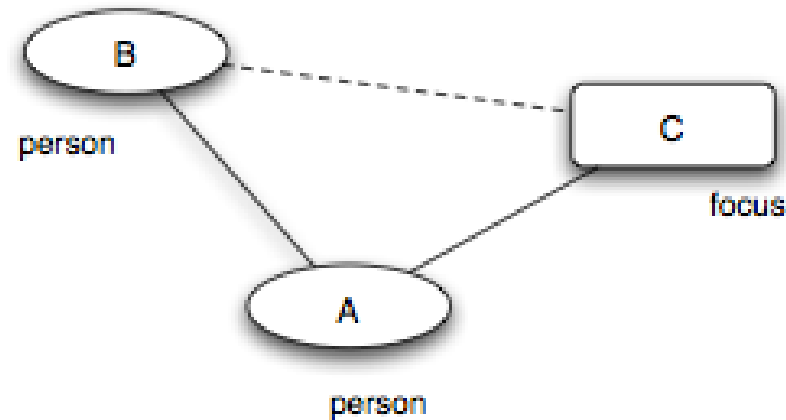
Three Closure Processes



(a) *Triadic closure*



(b) *Focal closure*



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. (Same mechanism as Lecture 03.)

Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. (*New — the focus mediates the introduction.*)

Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. (*New — the friend mediates membership.*)

Speaker notes

All three are mechanisms that turn structure into *more* structure. Triadic closure was L03's core concept; focal and membership closure are L05's additions. In the classroom figure, within-track friendships could arise through focal closure (shared labs and lectures bring STEM students together), and if an Arts student (A1) befriends a STEM student (S4) and later joins a STEM elective, that is membership closure. The distinction matters because focal and membership closure produce homophily-like patterns *without* any personal preference for similarity.

Closure Processes — Classroom Examples

Closure type	Open path	What closes?	Classroom example
Triadic	student → friend → student	A new friendship	"My lab partner introduces me to their friend."
Focal	student → elective → student	A new friendship	"We sit in the same statistics elective and start talking."
Membership	student → friend → club	A new membership	"My friend invites me to join the robotics club."

Only triadic closure needs a shared friend. Focal closure can create ties among strangers through repeated exposure in the same setting.

This table gives students a concrete classification handle before the quick check. Emphasize that focal closure is often enough to create observed homophily: people in the same track share many foci, so within-track ties become more likely even without an explicit preference for same-track friends.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

Speaker notes

Give them 3 minutes before revealing the fragments. (1) Focal closure — the elective is the shared focus. (2) Triadic closure — the new friendship forms via a shared friend (the advisor). (3) Membership closure — your friend introduces you to a focus you did not previously belong to.



Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
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1. Focal closure — the elective is the shared focus that brings them into contact.

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2. Triadic closure — the advisor is the shared friend that closes the triangle.

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2. Triadic closure — the advisor is the shared friend that closes the triangle.

3. Membership closure — your friend brings you into a focus you did not previously belong to.

Takeaway

Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

Online Link Formation

Guiding Question: What can large-scale online data teach us about how ties form over time?

Online platforms make link formation a measurable process at scale. Every new edge has a timestamp, and every preceding state is observable. This is what makes large-scale empirical tests of closure possible — and it is also where we finally get the longitudinal data that Modules 1–2 argued we need.

The Setting — Kossinets & Watts (2006)

Data. A large U.S. research university's internal email logs, ~~1 year~~, ~~**~~43,000 students/faculty/staff**, ~14 million messages — plus full course-enrollment records and organizational-affiliation rosters.

Why it matters. For the first time, all three closure types could be observed *with timestamps* on a single population: who emails whom, who shares which classes, who joins which clubs.

What was measured. At every time step, for each pair (u, v) *not yet connected*, count the number k of "closing paths": shared friends (triadic), shared foci (focal), or friend-mediated foci (membership). Then look at the probability that the pair connects in the next interval.

The output is a curve: $P(\text{new tie})$; vs. k .

Speaker notes

Three reasons to flag the setting before showing curves: (i) the population is large enough that very rare patterns (e.g., new ties between strangers sharing many foci) appear in statistically usable numbers; (ii) the timestamps make the analysis *forward-looking* — we look only at pairs not yet connected and ask whether they connect next; (iii) the same study generates all three curves on one platform, so the *shapes* can be compared directly. Students often forget that "online data" here means email logs from 2003–2004, not social-media data.



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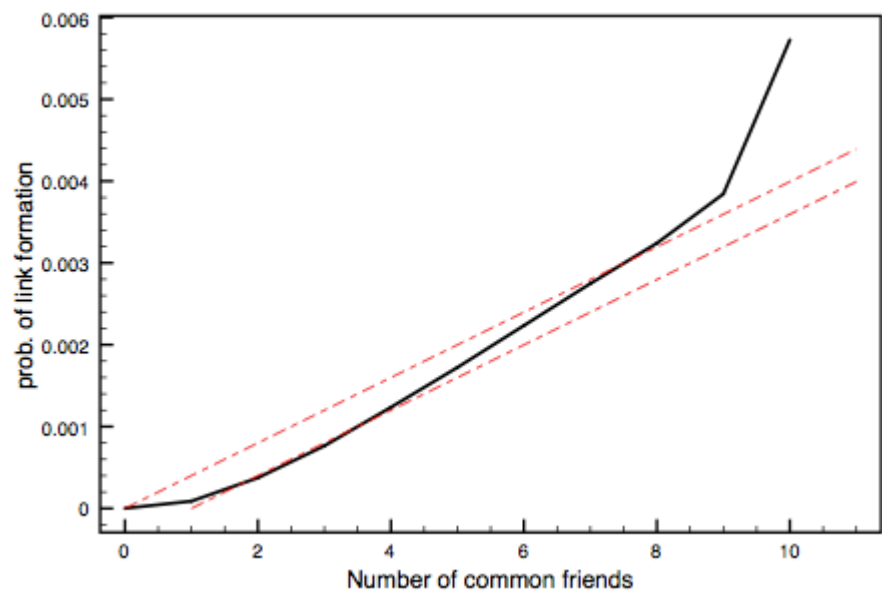
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This is the dataset behind the three closure-curve figures in Easley & Kleinberg Ch. 4. The timestamped data finally separates "*what existed at t* " from "*what formed by $t + 1$* " — the missing ingredient in homophily snapshots.

Triadic Closure at Scale



Source: Easley & Kleinberg 2010

Setup. Two people who have **never** emailed each other but share k mutual email contacts. What is the probability they exchange email within the next interval?

k shared friends	$P(\text{new tie})$
0	very low (baseline)
1	sharply higher
2	higher still
5+	curve flattens — saturation

Two readings.

1. **Mechanism.** Mutual friends create *opportunities*: introductions, shared

interests, social pressure to know each other.

2. **Diminishing returns.** The 1st shared friend matters far more than the 10th — mutual friends are *not* independent channels; they often belong to the same group.

The empirical curve is the central evidence for triadic closure at scale. Walk students through three landmarks: (i) the baseline rate at $k = 0$ — essentially how often two random strangers in the population start emailing — is very small; (ii) the *first* shared friend produces the biggest jump; (iii) the curve saturates. The saturation is itself the finding: it rules out a "simple counting" model and points to redundancy among mutual friends. This redundancy connects directly to L04's embeddedness idea.

A Toy Model for the Shape

If each shared contact independently offers an introduction opportunity with probability q , then the probability of closure given k shared contacts is:

$$P(\text{new tie} \mid k \text{ shared}) = 1 - (1 - q)^k$$

This *saturating* curve rises steeply at first and flattens as k grows — matching the empirical shape qualitatively. But real curves often flatten *faster* than this model predicts, suggesting that shared contacts are not fully independent.

Speaker notes

The $(1 - (1 - q)^k)$ model is a useful baseline because it is the simplest model that produces saturation. The fact that real data flattens *faster* than the independent model is itself informative — it suggests redundancy among shared contacts, consistent with the embeddedness idea from L04. Students who have seen survival analysis will recognize this as a "complement of a geometric-tail" form.

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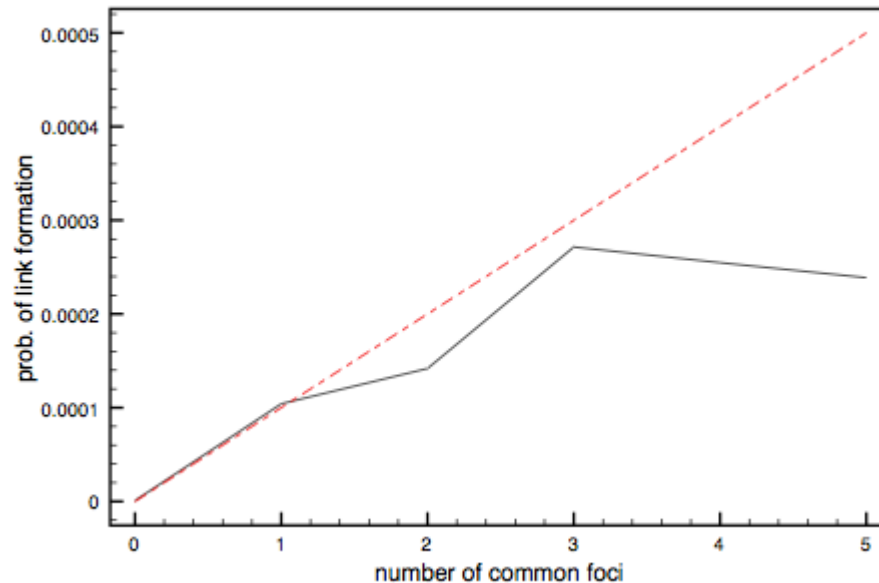
Example with $q = 0.20$:

Shared contacts k	1	2	3	5
$P(\text{new tie} \mid k)$	0.20	0.36	0.49	0.67

The first shared contact adds the most. Later shared contacts add less because many opportunities are redundant.



Focal Closure at Scale



Source: Easley & Kleinberg 2010

Setup. Two people who have **never** emailed each other but co-enroll in k classes during the same semester. What is the probability they exchange email within the term?

Result. Same qualitative shape as triadic closure: sharp rise from $k = 0$, then saturation. Sharing one class is already informative; 3+ classes produces a much higher closure rate.

Why this matters.

- Focal closure runs **without any shared friend**. Strangers in the same context form ties through repeated exposure alone.
- This is the mechanism that turns "people share classes" into "people share friendships" — and it produces homophily-like patterns even when

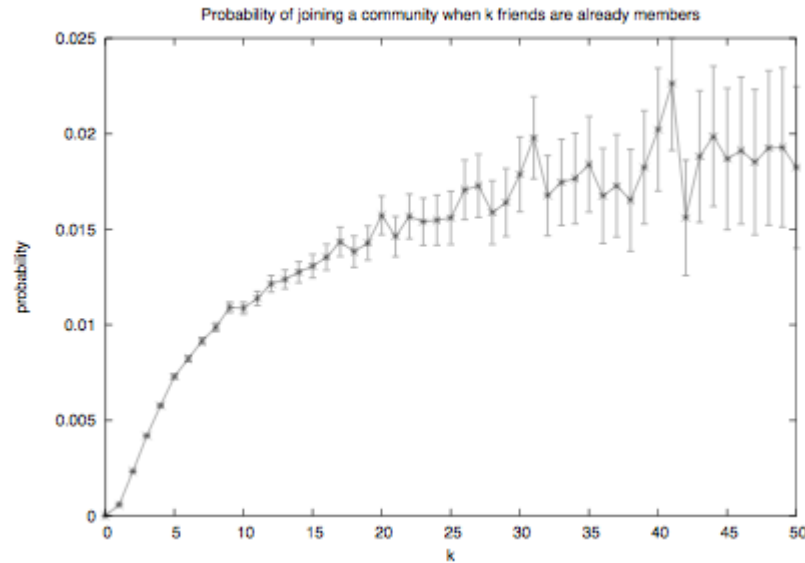
there is **no preference for similarity at all**.

- It is L05's main answer to *Question 3* from the opening slide ("could shared context explain the clustering?"). Often, yes.

Speaker notes

Focal closure is L05's distinctive concept (triadic was L03's). The empirical signature is the same saturating curve, but the mechanism is different: instead of introductions through mutual friends, the focus itself provides repeated exposure. The teaching link to homophily is direct: if STEM students share many classes, focal closure alone can generate a homophily-rich within-STEM friendship network without any selection effect. This is exactly the "third explanation" (context) that Module 2 introduced — and the affiliation network model from Module 3 is the formal representation.

Membership Closure at Scale



Source: Easley & Kleinberg 2010

Setup. A person is **not yet a member** of a focus (course, club, group). How does the probability of joining depend on the number of existing members who are already their friends?

Result. Same saturating shape. One friend already in the focus is a strong predictor of joining; additional friends compound — but with diminishing returns.

The flipped direction.

- Triadic and focal closure close *person-person* ties.
- Membership closure closes a *person-focus* tie — your friends pull you **into** the contexts they already inhabit.
- This is how **social networks shape affiliations**, not just the other way around. Friendships and foci co-

Speaker notes

Membership closure flips the bipartite arrow: instead of foci producing person–person ties (focal closure), person–person ties produce focus-memberships. This is the third closure mechanism and the one most often missed in introductory treatments. In the classroom example, if an Arts student A1 has multiple STEM friends, A1 is more likely to *take* a STEM elective. This is how friendships drive affiliations, completing the co-evolution loop between the network and the foci.

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Steep initial rise, then flattening. (2) $q \approx 0.20$ at $k = 1$. (3) The model overpredicts at $k = 2$; the redundancy among shared friends explains the gap. Connect this back to L04's embeddedness idea: shared friends are often embedded in the same dense cluster, so each additional shared friend gives less *new* opportunity.

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1. Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%) — a **saturating** curve.
2. At $k = 1$: $1 - (1 - q) = q$, so $q \approx 0.20$.

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1. Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%) — a **saturating** curve.
2. At $k = 1$: $1 - (1 - q) = q$, so $q \approx 0.20$.
3. The model predicts $1 - (1 - 0.20)^2 = 0.36$ but we observe **0.30** — a shortfall. Shared friends are *correlated* (typically embedded in the same group), so they provide less independent opportunity than the model assumes. Diminishing returns are *steeper* than independence predicts.

Takeaway

Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rather than independent.

From Local Preferences to Global Segregation

Guiding Question: Can weak local preferences generate strong global segregation?

Speaker notes

This module answers Question 4 from the opening slide: "Would mild preferences suffice to create the sharp pattern we see?" Schelling's threshold model shows that the answer is yes — and the mechanism is iterated local adjustment, not central planning.



Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

The formal model is simple enough to simulate in a few lines of code, yet its output is dramatically disproportionate to its input. The agent asks for 33% same-type neighbors; the iterated dynamics deliver 90%+ segregation. No agent *wanted* the global outcome; it emerged from the accumulation of individually mild adjustments.

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Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

Schelling — A Local Rule

Let $\tau = 1/3$. An agent compares its local neighborhood to that threshold.

Same-type neighbors	Total neighbors	Same-type fraction	Status
2	8	0.25	Unsatisfied: move or rewire
3	8	0.375	Satisfied: stay
4	8	0.50	Satisfied: stay

The rule is mild: the agent does **not** require a majority of similar neighbors. Strong segregation emerges because many mild local moves are repeated.

Speaker notes

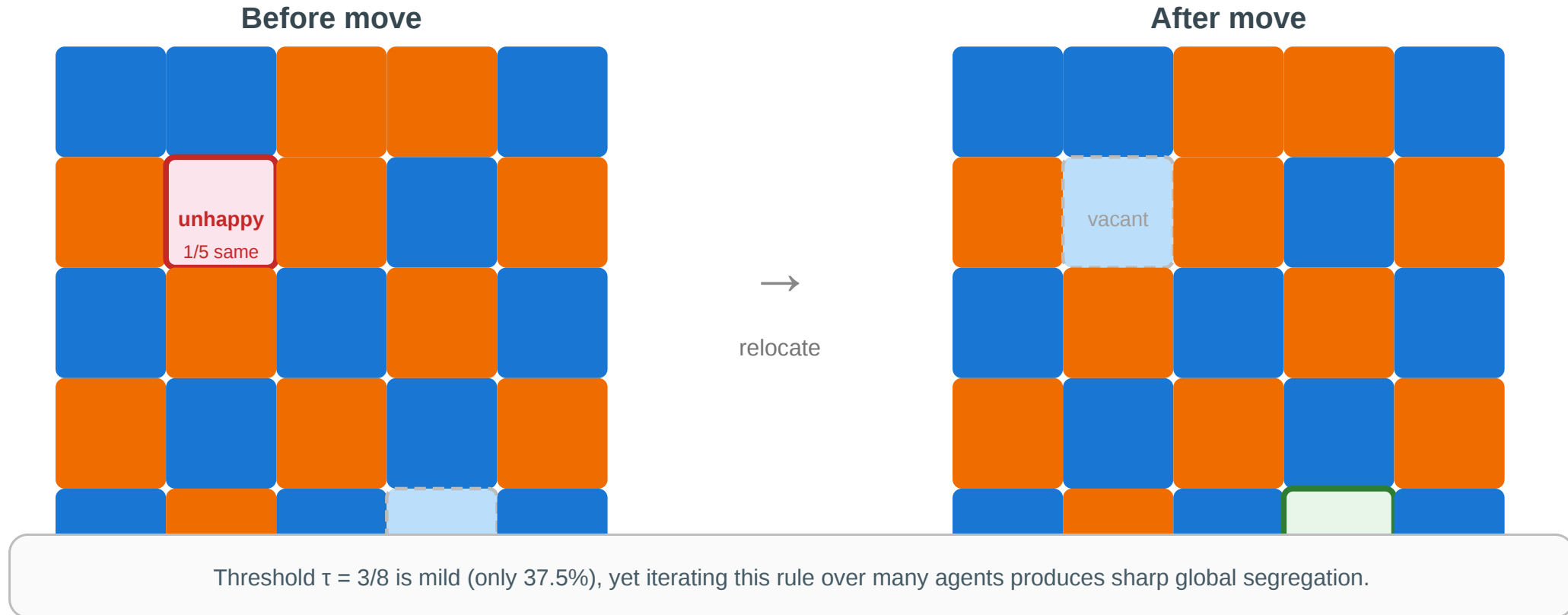
This numerical slide makes the threshold rule operational. Students should see that the model does not assume extremist preferences. With a threshold of one third, an agent with 3 of 8 same-type neighbors stays. The global segregation is an emergent outcome of iteration, not a direct expression of each agent's desired final map.



One Step of the Model

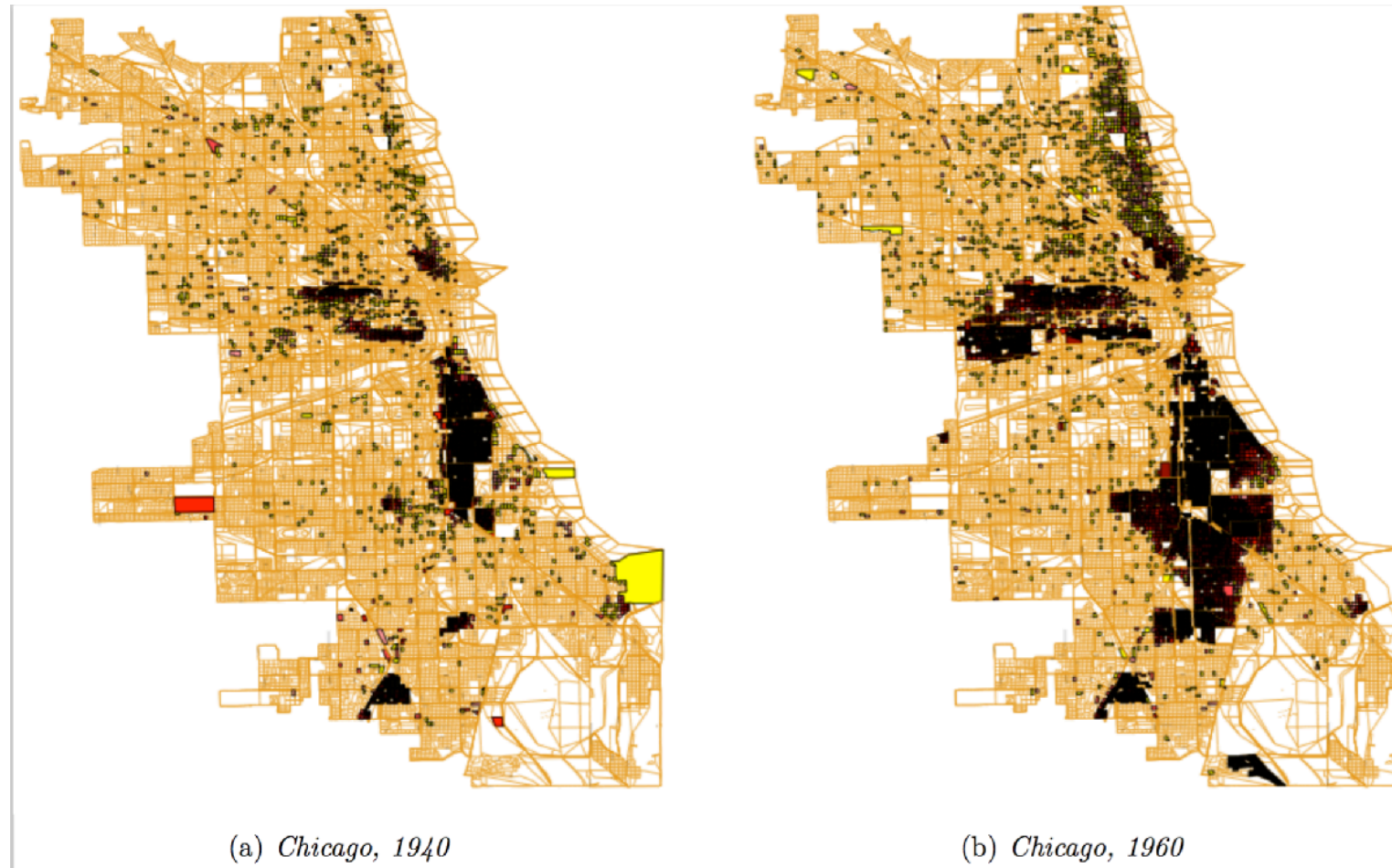
Schelling's threshold model — one step

Each agent wants at least $\tau = 3/8$ of its neighbors to be same-type. Unhappy agents relocate.



The blue agent at (1,1) has 1/5 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

Empirical Motivation



Source: Easley & Kleinberg 2010

Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

Speaker notes

The empirical map raises the puzzle: how can boundaries be *this* sharp if individual preferences are not extreme? Schelling's model answers: iterated local moves amplify mild preferences into sharp global patterns. The same logic applies to online homophily, recommendation systems, and network rewiring.



From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

Speaker notes

This is the network-science translation of Schelling. The move from a spatial grid to a graph model changes the topology but preserves the core insight: mild local rules plus iterated updates produce sharp global segregation. The recommendation-algorithm connection brings the 1971 model into the 2020s: algorithms now run the Schelling update at machine speed, with consequences for filter bubbles, echo chambers, and political polarization.



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Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

Speaker notes

Give them 4 minutes before revealing the fragments. (1) Network neighbors are graph-adjacent nodes. (2) "Moving" becomes rewiring — dropping ties to dissimilar contacts and adding ties to similar ones. (3) The platform acts as a Schelling updater: it identifies "unsatisfied" tie configurations (low similarity) and offers replacements (higher similarity). It amplifies homophily by accelerating the rewiring process.

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1. Network neighbor: a graph-adjacent node — your immediate contacts.

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- 1. Network neighbor:** a graph-adjacent node — your immediate contacts.
- 2. Moving:** rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.
- 3. The platform is an automated Schelling rewirer.** It surfaces similar people — equivalent to offering each agent a move toward a higher same-type fraction. This *amplifies* homophily: the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Takeaway

Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — The Three Gaps

Across Lectures 03–05, the course has exposed three kinds of gap between what we *want to know* about a network and what the data *actually shows*:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies (clustering, overlap)
L04	Computational	Optimal partition ($\max Q$)	NP-hard	Heuristics (GN, Louvain, Leiden)
L05	Causal	Mechanism (selection vs. socialization)	Not identifiable from snapshots	Longitudinal data, affiliation models

Every concept in the three lectures occupies one cell of this table.

Looking Ahead: Lecture 06

If signs (positive and negative) — not just connection strengths — matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy

relations into a graph-theoretic claim with measurable predictions.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Speaker notes

The three-gap table closes the L03–L05 arc. The computational gap and the causal gap are fundamentally different in character: the first is about algorithms, the second is about study design. Both are present in any real network analysis, and the responsible analyst must address both. Lecture 06 opens a new thread — signed graphs and structural balance — which introduces yet another kind of constraint: not computational or causal but *logical* (can the sign pattern be consistent across all triangles?).



Image Credits & References

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